Online Appendix for: The Fertility Dividend: Family Size and the Decline in Child Poverty

November 2026

This Appendix contains robustness analyses, extended tables, and supporting figures. Appendix A provides additional identification and decomposition details beyond those in the main text. Appendix B reports robustness checks varying the transport method, tiebreaker, time period, conditioning set, and poverty measure. Appendix C provides extended twin-birth IV regression tables, including first-stage estimates and OLS comparisons. Appendix D provides supporting figures for the long-run analysis. Appendix E provides supplementary detail on data sources and sample construction.

A Identification and Decomposition

A.1 Rank-Preserving Distributional Transport

Within demographic cells defined by marital status, I construct counterfactual poverty rates by replacing the distribution of co-resident children in one year with that of another while holding resources and other family characteristics fixed. Let C_{it} denote the number of co-resident children for mother i in year t. In the base year, I rank mothers within each cell by C_{it} and compute their weighted percentile rank. Because C_{it} is discrete, many mothers share the same value. In the main specification, I therefore break ties by assigning a random order within each family-size category. This yields a unique percentile rank $p_i \in (0, 1)$ for each mother.

I then replace each base-year mother's number of co-resident children with the counterfactual number of co-resident children in the target year at the same CDF rank. Specifically, within the same cell I form the target year's weighted empirical CDF of co-resident children, $F_c^{\text{tgt}}(\cdot)$, and assign mother i the smallest integer c such that $F_c^{\text{tgt}}(c) \geq p_i$.

Intuitively, a mother at percentile p of the distribution of number of children in the base year is reassigned the number of children at percentile p of the target year's distribution, preserving the joint ordering of fertility propensities within the cell even as the marginal distribution shifts.

Given the counterfactual number of children C_i^{cf}, I recompute poverty holding the mother's observed resources fixed. I adjust the SPM poverty threshold by the ratio of equivalence-scale factors implied by (A_i, C_i^{cf}) versus (A_i, C_{it}) (see below), and then classify the mother as poor if family resources fall below the counterfactual threshold. Finally, I calculate the average child poverty rate using counterfactual child weights equal to the mother weight multiplied by C_i^{cf}. The resulting counterfactual poverty rate isolates the direct resource-dilution effect of changes in family size within cells, holding the distribution of resources fixed.

This procedure adapts the rank-preserving distributional transport approach of Novosad et al. (2022). Relative to reweighting methods (DiNardo et al. 1996), the advantage is that it preserves the joint distribution of family size and resources within cells rather than matching only on conditional means. This distinction matters because poverty is determined by applying a family-specific poverty threshold to the distribution of family income. Shifting family size can move families across the poverty line even when average resources are unchanged.

A.2 Identifying Assumption

The counterfactual assignment of fertility relies on a rank-stability (or copula-stability) assumption for fertility within demographic cells. Let θ_i denote an unobserved, time-invariant

"fertility propensity" for mother i, and let $X_i = x$ index the cell (e.g., marital status). The identifying assumption is that the conditional distribution of θ_i is stable over time:

$$\Pr(\theta_i \leq \theta \mid X_i = x, t = t_0) = \Pr(\theta_i \leq \theta \mid X_i = x, t = t_1)$$

for all θ and x. Under this assumption, a mother at percentile p of the family-size distribution within cell x in year t_0 would also be at percentile p within cell x in year t_1, so transporting fertility by matching percentiles recovers the counterfactual family-size distribution.

This assumption is different from (and typically weaker than) selection-on-observables assumptions used to identify causal effects via reweighting or propensity score matching. In particular, it does not require fertility to be independent of potential outcomes conditional on X. High-fertility mothers may have systematically lower resources, so long as the relative ranking of fertility propensities within X is preserved as the marginal distribution of fertility shifts over time. The assumption could be violated if there were changes in fertility preferences specific to certain income groups, or changes in the relative population shares of high or low fertility groups that reordered fertility preferences within cells. The assessment of the robustness of the main estimates to alternative rankings within demographic cells and tie-breaking rules is intended to address some potential threats to identification.

A.3 Threshold Adjustment

When counterfactual fertility changes a mother's number of children, I adjust the SPM poverty threshold to reflect the counterfactual family size while preserving all other components of the original threshold (e.g., geographic cost-of-living adjustments). The SPM

scales thresholds using the Betson three-parameter equivalence scale (Betson 1996):

$$\text{scale}(A, C) = \begin{cases} (A + 0.5\,C)^{0.7} & \text{if } A \geq 2, \\ (1.3 + 0.5\,C)^{0.7} & \text{if } A = 1 \text{ (single-adult unit with children).} \end{cases}$$

Here A is the number of adults in the SPM resource unit and C is the number of co-resident children. In the fertility transport, the number of adults in the SPM unit (A) and marital status are held fixed; only the number of children (C) changes. Let $\text{threshold}_{\text{orig}}$ denote the observed SPM threshold for the mother in the base year. The counterfactual threshold is constructed by rescaling the observed threshold by the ratio of equivalence-scale factors implied by the counterfactual versus original number of children:

$$\text{threshold}_{\text{cf}} = \text{threshold}_{\text{orig}} \times \frac{\text{scale}(A, C_{\text{cf}})}{\text{scale}(A, C_{\text{orig}})}.$$

This adjustment changes only the number-of-children component of the threshold.

A.4 Shapley Value Decomposition

I implement a two-stage Shapley value decomposition (Shapley 1953; Shorrocks 2013; Young 1985) that (i) attributes changes in total child poverty to fertility and marriage, and then (ii) decomposes the remaining change into market income versus taxes and transfers. The Shapley value averages each factor's marginal contribution over alternative orderings, producing order-independent components that sum to the total change.

Stage 1: Fertility and marriage. Starting from the reference-year population, I measure counterfactual child poverty under the above fertility and marriage changes applied in both orderings. I work on a mother-level dataset and aggregate to the child level using weights $\omega_i \times C_i^{\text{cur}}$, where ω_i is a running weight initialized to the SPM person weight and C_i^{cur} is the current (possibly counterfactual) number of children. In the main specifica-

tion, the fertility update transports family size within marriage cells: using the target-year weighted CDF $F_m^{t_1}(c) = \Pr(C \le c \mid M = m, t = t_1)$ and each mother's within-cell weighted percentile

$$p_i = \frac{\sum_{j \le i} \omega_j}{\sum_{j \in \text{cell}(i)} \omega_j},$$

with ties in discrete family size broken by a uniform draw, I assign the counterfactual family size

$$C_i^{\text{cur}} \leftarrow \min\{c : F_{M_i}^{t_1}(c) \ge p_i\},$$

update the SPM threshold by the implied equivalence-scale ratio (Section A.3), and recompute poverty. The marriage update holds marital status fixed and reweights observations to the target-year child-weighted marriage share μ^{t_1},

$$\omega_i \leftarrow \omega_i \times \begin{cases} \mu^{t_1}/\mu^{t_0} & \text{if } M_i^{\text{cur}} = 1, \\ (1-\mu^{t_1})/(1-\mu^{t_0}) & \text{if } M_i^{\text{cur}} = 0, \end{cases}$$

matching the target-year marriage composition while holding within-group distributions fixed.

Stage 2: Market income versus taxes and transfers. Stage 1 yields Shapley contributions of fertility and marriage to the change in total child poverty, along with counterfactual poverty rates after applying these factors. With the fertility and marriage contributions in hand, the remaining change is allocated between market income and taxes and transfers using the fact that total poverty is computed from total resources and market poverty is computed from pre-tax, pre-transfer resources against the same (size-adjusted) threshold. Changes in market poverty capture shifts in market resources, while the gap between total and market poverty captures the incremental poverty reduction due to taxes and transfers. I treat market income and taxes/transfers as the two Stage-2 factors and average each factor's marginal contribution over both orderings—market poverty first in

one, the tax-and-transfer gap first in the other—to obtain order-independent components that sum exactly to the Stage-1 residual.

A.5 SPM Decade Decomposition

Table A1: Decade-by-decade Shapley decomposition of child poverty change

Period	Total Change	Fertility	Marriage Decline	Market Income	Taxes/ Transfers
1967-1980	-4.84	-4.26	3.59	-2.39	-1.78
1980-1990	1.19	-0.58	1.41	0.68	-0.32
1990-2000	-8.15	-0.63	0.74	-5.83	-2.44
2000-2010	-0.45	-0.05	0.89	4.81	-6.11
2010-2024	-7.34	0.08	-0.23	-8.04	0.85
Cumulative	-19.59	-5.43	6.40	-10.77	-9.79

Notes: All values in percentage points. Each row decomposes the change in anchored SPM child poverty within that period using Shapley values. Fertility uses rank-preserving CDF transport; Marriage uses IPW reweighting; Market income and Taxes/Transfers use a nested Stage 2 Shapley. Negative values indicate poverty reduction. Source: CPS ASEC, 1967–2024.

The main text presents the cumulative 1967–2024 decomposition (Figure 2) and highlights key decade-level patterns (Figure 3). Table A1 provides the data underlying those figures. The fertility effect was concentrated in 1967–1980 (−4.3 pp), when average family size fell rapidly. After 1980, fertility's marginal contribution was relatively smaller. Market income growth was a more significant driver of declining poverty in the 1990s and 2010s and changes in taxes and transfers were more important in the 2000s. The cumulative sum across decades (−19.6 pp) closely approximates the single-period 1967–2024 decomposition total (−19.8 pp). The discrepancy arises because Shapley contributions are computed separately for each decade, and the decomposition is not additively path-independent in levels with a nonlinear poverty statistic.

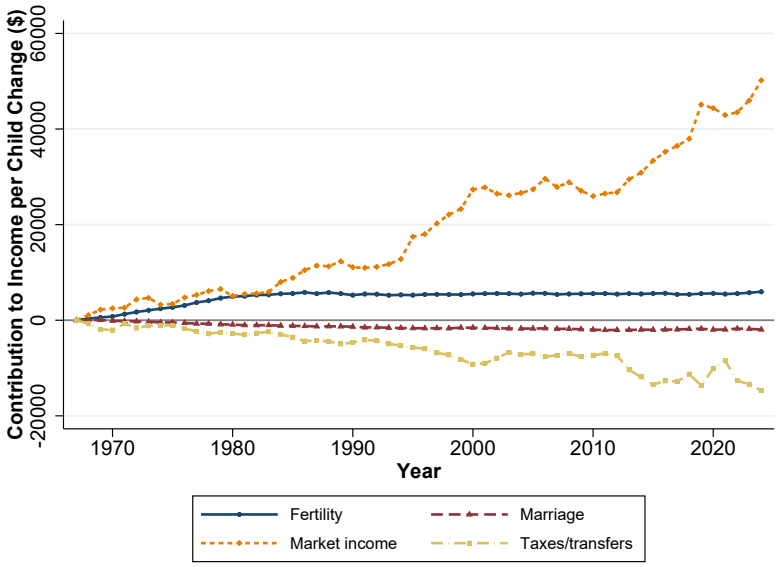


Figure A1: Decomposition of mean real income per child, 1967–2024: market income growth dominates

Notes: Cumulative Shapley contributions to the change in mean real income per child, 1967–2024. Income per child is SPM resources divided by the number of children in the resource unit, aggregated using child weights (mother survey weight × number of children) and deflated to 2024 dollars using the PCE price index. Source: CPS ASEC, Historical SPM.

A.6 Income per Child

The main text decomposes changes in poverty rates. The same framework can be applied to mean real income per child (SPM resources per family divided by the number of children, PCE-deflated to 2024 dollars). Figure A1 plots the cumulative Shapley contributions to the change in mean income per child. Market income growth is the most important factor (+$50,000), and declining family size contributed about +$6,000 per child.

The net contribution of taxes and transfers is negative in this analysis. This is because income per child reflects the average net financial position of families with children. Because the SPM defines resources as market income plus transfers minus taxes and necessary expenses (including work expenses and medical out-of-pocket costs), rising tax liabilities and required expenses can reduce families' average net incomes even as targeted programs such as the EITC, SNAP, and housing assistance reduce poverty at the poverty-threshold margin.

B Robustness of Main Decomposition Results

The main decomposition estimate of -5.4 pp relies on three methodological choices: the conditioning set (marriage cells), the tiebreaker for within-cell ranking (random), and the assumption that the fertility effect is stable across sub-periods.

B.1 Transport Method and Conditioning Set Variations

The main specification (A) transports the distribution of family size within marital-status cells using the rank-preserving transport described above. I assess three alternatives that vary the conditioning set and assumptions regarding fertility timing. Method B performs the transport ignoring marital status. In Method C, within marital-status cells, I transport exact age at first birth (AFB) to match the target-year AFB distribution and then transport family size within the finer cells defined by marital status $\times$ AFB quintile. The idea is that this matches first on AFB and then on fertility rank. Finally, Method A+IPW implements a timing-adjusted variant of the main approach. Before transporting family size within marriage cells, I reweight the reference-year sample so that the distribution of age at first birth matches that of the target year. Then I transport fertility within the reweighted population. The difference between A+IPW and A provides an accounting measure of the role of shifts in fertility timing.

Many families have the same number of children, and the rank-preserving match is not unique within these discrete family sizes. I therefore adopt tie-breaking rules based on different assumptions about how latent fertility is distributed within cells. Following the partial-identification logic of Manski (2003), these generate a range of counterfactual rankings within discrete sizes.

The main specification breaks ties at random, treating mothers within a family-size category as exchangeable. To bound the role of different approaches to breaking ties, I assess the consequence of instead using income. Sorting ties by ascending income assigns

the largest counterfactual family sizes to the lowest-income mothers. Because poverty falls with resources, this maximizes the implied poverty impact (an upper bound). Sorting by descending income provides a lower bound. The two rules bracket the range of rank–income relationships within family size.

An alternative assumption is to preserve the fertility-income rank that prevails generally: a Gaussian copula mapping income and family-size percentile ranks to normal scores. This produces a copula whose correlation equals the within-cell sample rank correlation. In this case, fertility and income ranks are monotonically but imperfectly related.

Table 1 (main text) reports the results. Across all 16 specifications, the estimated fertility contribution ranges from -2.3 to -8.5 percentage points. The random-tie estimates—the main specification within each method—cluster between -4.9 and -6.1 pp, bracketing the baseline Shapley estimate of -5.4 pp. Methods A and C yield similar estimates, indicating that conditioning on age at first birth adds little beyond conditioning on marital status. Method A+IPW produces a larger estimated contribution than A alone: under the random tie-breaker the difference is about -0.8 pp (and -1.0 pp under the copula), reflecting the incremental change associated with shifts in the age-at-first-birth distribution in this accounting framework.

In additional checks, I expand the transport cells from marital status alone (2 cells) to race $\times$ marital status (4 cells: white married, white unmarried, nonwhite married, nonwhite unmarried). Adding race—a predetermined covariate—yields a slightly larger estimated fertility effect (-5.7 pp versus -5.0 pp under the single-transport comparison), indicating that the main specification is not inflated by compositional changes across racial groups.

B.2 Sub-Period Stability

Table B1 repeats the transport comparison separately for 1967–1990 and 1990–2024. Under the main specification (A, random tie-breaker), fertility reduces child poverty by 4.9 pp

in 1967–1990 but by only −0.8 pp after 1990, consistent with the slowdown in declines in completed family size over the later period. In contrast, the timing-adjusted variant (A+IPW) delivers a larger estimate in 1990–2024 (−3.0 pp versus −0.8 pp for A). This pattern indicates that after 1990, changes in fertility timing (as reflected in the age-at-first-birth distribution) played a larger role in poverty reduction than reductions in family size.

Table B1: Fertility effect by sub-period (pp, random tiebreaker)

Transport Method	1967–2024	1967–1990	1990–2024
A: Within marriage	−5.3	−4.9	−0.8
B: Marginal	−5.0	−4.6	−0.6
C: Exact AFB → fert	−4.8	−4.6	−0.4
A+IPW: Marr + AFB IPW	−6.1	−5.8	−3.0

Notes: Total poverty change: −20.1 pp (1967–2024), −4.0 pp (1967–1990), −16.1 pp (1990–2024). Cells report single-transport fertility effects; the main-specification 1967–2024 value (A, −5.3 pp) corresponds to the −5.4 pp Shapley fertility contribution in the text, which adds the marriage-ordering interaction. Source: CPS ASEC.

B.3 Robustness to Fertility Timing: Age-at-First-Birth Reweighting

Method A+IPW adjusts the reference-year composition of mothers to match the target year's age-at-first-birth distribution before applying the baseline fertility transport. This isolates how much of the change in child poverty is attributable to shifts in birth timing holding fixed the reference-year relationship between timing, resources, and poverty. This adjustment is not intended to be an estimate of the causal effect of contraception or the delay in the timing of fertility on education, labor supply, or earnings. By construction, the reweighting preserves the reference-year age–earnings and age–poverty profiles, and the contemporaneous institutional environment, so it cannot capture the broader behavioral changes emphasized in the contraception literature.

B.4 Robustness to Alternative Poverty Measures

The main decomposition uses SPM poverty, which adjusts thresholds for family size via the Betson three-parameter equivalence scale. To demonstrate that the fertility contribution is not an artifact of this particular scale, Table B2 recomputes the fertility-transport contribution under six alternative poverty definitions. Each row applies the same rank-preserving transport of family size (within marital-status cells, random tiebreaker, 1967→2024) and re-estimates poverty using each measure. The square-root scale replaces the Betson adjustment with a $\sqrt{N}$ equivalence factor; the per-capita measure assumes no economies of scale, so that each family member claims an equal share of resources; the Lazear and Michael (1980) and Nelson (1988) rows apply the two equivalence scales estimated directly from household expenditure data—these use the largest economies of scale in the literature (Fernández-Villaverde and Krueger 2007), with scale values for sizes 1–5 taken from FVK's Table 1 and sizes 6 and above extrapolated linearly using the last observed per-person increment; deep poverty uses the Betson threshold at 50%; and the fixed per-child cutoff classifies a child as poor if real family resources per child fall below $20,000 (2024 dollars), independent of any equivalence scale.

Table B2: Sensitivity of the fertility contribution to alternative threshold-scaling rules

Poverty measure	Baseline	CF rate	Effect (pp)
SPM (Betson scale)	29.4	24.4	−5.0
Square-root scale	29.4	24.5	−4.9
Per-capita (no economies of scale)	29.4	21.2	−8.2
Lazear–Michael (1980)	29.4	23.4	−6.0
Nelson (1988)	29.4	26.2	−3.2
Deep poverty (50% of SPM threshold)	8.5	7.0	−1.5
Resources per child < $20k (2024$)	64.6	47.8	−16.8

Notes: Each row applies the same rank-preserving fertility transport (within marital-status cells, random tiebreaker, 1967→2024) and recomputes poverty under a different definition. The square-root and per-capita scales rescale the observed SPM threshold by the ratio of alternative equivalence factors implied by the counterfactual versus original number of children. Baseline poverty rates are identical across the first five rows because each classifies families using the same observed 1967 SPM threshold; differences arise only under counterfactual family sizes, when the threshold is rescaled by the ratio of equivalence factors. The fixed per-child cutoff is scale-free: it classifies a child as poor if real family resources divided by the number of children fall below $20,000 in 2024 dollars. Source: CPS ASEC, 1967–2024.

The fertility contribution to poverty change is robust across all measures. The Betson and square-root scales yield nearly identical estimates (−5.0 and −4.9 pp). The consumption-based scales of Lazear and Michael (1980) and Nelson (1988) bracket this benchmark (−6.0 and −3.2 pp respectively). The per-capita measure, which assumes no economies of scale, produces a larger effect (−8.2 pp). The contribution of fertility also extends to deep poverty (−1.5 pp from an 8.5% baseline) and to the scale-free per-child resource measure (−16.8 pp). Across these measures, the family-size channel remains negative and of similar magnitude. The −5.0 pp single-transport estimate is not an artifact of the SPM's scale.

C Twin Birth Instrumental Variables

This appendix provides the full specification and results for the twin-birth IV analysis summarized in the main text.

C.1 Sample and Specification

The analysis uses two IPUMS-USA samples (Ruggles et al. 2023): the 1960 Decennial Census and pooled ACS microdata for 2015–2024, which are grouped into a single "modern" sample. The unit of observation in the IV regressions is the mother: I restrict to children under 18 who can be linked to a mother, construct completed (or near-completed) fertility measures at the mother level, and keep one observation per mother. The IV sample is restricted to mothers aged 30 and older.

Twin births are identified as instances in which two children linked to the same mother share a birth year and birth quarter. Triplets and higher-order multiples are excluded by dropping pregnancy–mother cells with more than two births.

C.2 Parity-Threshold 2SLS Design

I estimate parity-specific local average treatment effects at margins $n \in \{1, \ldots, 6\}$. Let Twin_{in} indicate that mother i has a twin birth at pregnancy n, and let $\text{Has}_{i,n+1}$ indicate that mother i has at least $n+1$ co-resident children (completed family size $\geq n+1$). For each margin n, the sample is restricted to mothers with at least n pregnancies.

First stage.

$$\text{Has}_{i,n+1} = \alpha_{0n} + \alpha_{1n}\,\text{Twin}_{in} + \mathbf{X}_i'\beta_n + \varepsilon_{in}.$$

Second stage.

$$Y_i = \pi_{0n} + \pi_{1n}\,\widehat{\text{Has}}_{i,n+1} + \mathbf{X}_i'\theta_n + u_{in},$$

where Y_i is an outcome measured at the mother/family level and $\mathbf{X}_i$ includes fixed effects for maternal age, education, marital status, and age at first birth (proxied by the minimum maternal age-at-birth observed among co-resident children). In the modern sample, specifications also include survey-year fixed effects. Regressions are weighted by IPUMS person weights and use heteroskedasticity-robust standard errors.

Outcomes. The estimates use three behavioral resource outcomes for which the contrast between OLS and IV is informative about selection: (i) an indicator for child poverty under the Official Poverty Measure (OPM, poverty status ≤ 100); (ii) log total family income; and (iii) log earnings per child, defined as log family earnings divided by the number of children. A fourth outcome, the log official poverty threshold, is mechanical—a deterministic function of family size and composition—and measures the change in the family's needs standard induced by an additional child; it is reported in Panel D of Table 2 in the main text and enters the resources-versus-needs comparison ($\Delta \log R$ versus $\Delta \log T$) that frames the twin-birth IV analysis. Tables C2 and C3 report the full OLS and IV estimates for outcomes (i)–(iii); two final subsections add robustness to the age-window restriction

and twin-IV estimates for four mother-level labor-supply outcomes.

C.3 First Stage

Table C1 reports first-stage estimates. Twin births at pregnancy n increase the probability of having at least n+1 children by 0.36–0.71 (depending on the parity margin), with robust first-stage F-statistics exceeding 3,000 at every margin in both samples. First-stage coefficients are smaller in 1960, consistent with higher baseline fertility and greater scope for subsequent childbearing—and thus partial offsetting of the twin-induced parity shift—in the earlier period.

Table C1: First stage: effect of twin birth on crossing parity thresholds

	Parity Transition					
	1→2	2→3	3→4	4→5	5→6	6→7
Panel A: ACS 2015–2024						
Twin birth	0.486	0.670	0.705	0.703	0.646	0.602
	(0.001)	(0.001)	(0.002)	(0.003)	(0.005)	(0.010)
F-statistic	135,503	214,311	165,717	71,754	15,479	3,293
Observations	2,644,308	1,511,086	507,798	146,809	42,508	14,887
Panel B: 1960 Census						
Twin birth	0.357	0.463	0.501	0.514	0.493	0.488
	(0.002)	(0.002)	(0.002)	(0.003)	(0.004)	(0.005)
F-statistic	24,867	64,110	63,004	35,397	18,295	10,840
Observations	921,213	622,001	338,051	165,357	77,334	37,721

Notes: Each column reports the first-stage effect of a twin birth at pregnancy n on an indicator for having at least $n+1$ children (i.e., crossing the $n \rightarrow n+1$ parity threshold). Panel A reports estimates for the pooled ACS 2015–2024 sample; Panel B for the 1960 Census. The reported F-statistic is the robust first-stage Wald statistic for the excluded instrument, computed as $(\widehat{\alpha}_{1n}/\widehat{\text{se}}(\widehat{\alpha}_{1n}))^2$ from the first-stage regression. Unless noted otherwise, all IV tables in this appendix weight by IPUMS person weights, report heteroskedasticity-robust standard errors in parentheses and continuous p-values in brackets, restrict the sample to mothers aged 30+ with at least n pregnancies, and include the controls described in Appendix C.2.

C.4 OLS and IV Estimates

Table 2 in the main text summarizes the IV estimates across both samples; Tables C2 and C3 report the full results, including OLS comparisons, for the modern ACS and the

1960 Census respectively. The pattern is consistent across samples and is the signature of resource dilution: parity transitions raise child poverty (Panel A) and leave total family income (Panel B) essentially unchanged at higher parities, so earnings per child (Panel C) fall close to proportionally with the added child. Total income does not rise to accommodate the additional child—if anything maternal labor supply falls modestly (Appendix C.7)—and so the mechanical threshold increase (Table 2, Panel D) binds and child poverty rises.

Table C2: IV estimates: effect of crossing a parity threshold, ACS 2015–2024

	Parity Transition					
	1→2	2→3	3→4	4→5	5→6	6→7
Panel A: Child Poverty (OPM)						
OLS	0.053	0.068	0.069	0.080	0.071	0.082
	(0.001)	(0.001)	(0.002)	(0.003)	(0.006)	(0.010)
	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]
IV	0.031	0.037	0.043	0.038	0.079	0.065
	(0.004)	(0.004)	(0.009)	(0.016)	(0.037)	(0.063)
	[< .001]	[< .001]	[< .001]	[.017]	[.033]	[.306]
Panel B: Log Family Income						
OLS	0.005	−0.054	−0.050	−0.047	−0.024	−0.062
	(0.002)	(0.002)	(0.004)	(0.007)	(0.012)	(0.020)
	[.003]	[< .001]	[< .001]	[< .001]	[.040]	[.002]
IV	0.115	0.014	−0.011	0.039	−0.154	0.043
	(0.010)	(0.010)	(0.020)	(0.032)	(0.092)	(0.128)
	[< .001]	[.173]	[.595]	[.225]	[.093]	[.738]
Panel C: Log Earnings per Child						
OLS	−0.782	−0.555	−0.434	−0.394	−0.318	−0.312
	(0.002)	(0.002)	(0.004)	(0.008)	(0.014)	(0.024)
	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]
IV	−0.731	−0.459	−0.340	−0.272	−0.332	−0.234
	(0.011)	(0.011)	(0.021)	(0.040)	(0.092)	(0.140)
	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]	[.096]
Observations	2,644,308	1,511,086	507,798	146,809	42,508	14,887

Notes: Each column estimates the effect of crossing the indicated parity threshold. The endogenous regressor is an indicator for having at least $n+1$ children, instrumented with a twin birth at pregnancy n (e.g., the $2 \to 3$ column instruments having three or more children).

Table C3: IV estimates: effect of crossing a parity threshold, 1960 Census

	Parity Transition					
	1→2	2→3	3→4	4→5	5→6	6→7
Panel A: Child Poverty (OPM)						
OLS	0.106	0.127	0.147	0.177	0.172	0.134
	(0.001)	(0.001)	(0.001)	(0.002)	(0.003)	(0.005)
	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]
IV	0.099	0.084	0.095	0.126	0.143	0.158
	(0.011)	(0.010)	(0.013)	(0.018)	(0.026)	(0.033)
	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]
Panel B: Log Family Income						
OLS	−0.088	−0.102	−0.116	−0.128	−0.122	−0.094
	(0.002)	(0.002)	(0.002)	(0.004)	(0.006)	(0.009)
	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]
IV	0.017	−0.031	−0.026	−0.008	−0.060	−0.103
	(0.020)	(0.018)	(0.022)	(0.031)	(0.050)	(0.066)
	[.408]	[.082]	[.234]	[.784]	[.233]	[.119]
Panel C: Log Earnings per Child						
OLS	−1.054	−0.723	−0.579	−0.512	−0.451	−0.364
	(0.002)	(0.002)	(0.003)	(0.005)	(0.007)	(0.011)
	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]
IV	−1.111	−0.629	−0.490	−0.409	−0.271	−0.328
	(0.030)	(0.023)	(0.030)	(0.041)	(0.058)	(0.087)
	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]
Observations	921,213	622,001	338,051	165,357	77,334	37,721

Notes: Same specification as Table C2, estimated on the 1960 Census; the modern-sample specification includes survey-year fixed effects, which this single cross-section omits. Source: 1960 Census, IPUMS USA.

C.5 OLS versus IV and the Role of Selection

Comparing OLS and IV estimates is informative about selection. In both samples, OLS estimates generally exceed IV estimates, consistent with negative selection into larger families. IV estimates in 1960 are roughly twice as large as in the modern period, reflecting the greater density of families near the poverty threshold in 1960 rather than the higher baseline poverty rate per se.

A related concern is that assisted reproductive technology (ART) may induce selection at the $1 \rightarrow 2$ margin in modern data by increasing the incidence of twinning among higher-SES mothers. Consistent with this possibility, the $1 \rightarrow 2$ IV estimate for log family income is positive (0.115; Table C2). To the extent ART is correlated with income, it would attenuate the apparent relationship between twin births and poverty. In addition, the specifications control for maternal age, education, and marital status. Finally, the 1960 Census estimates predate ART entirely.

C.6 Age-Window Robustness

The main IV sample is restricted to mothers aged 30 and older to approximate completed fertility. This excludes younger mothers who account for a disproportionate share of poor families. This subsection shows that this restriction does not affect the results. Table C4 compares the modern-ACS IV estimates under the 30+ baseline with an alternative 18+ window that includes all mothers in the analysis age range. Coefficients are nearly identical across windows. The poverty effect at 30+ ranges from 3.1 to 7.9 pp across margins versus 3.4 to 6.5 pp at 18+. The log poverty threshold is essentially unchanged (because the threshold response is mechanical). And log family income remains close to zero in both windows. If anything, the poverty effects are marginally larger at low parities when younger mothers are included, consistent with their higher baseline poverty risk.

Table C4: Twin-IV estimates by age window: modern ACS, mothers 30+ vs. 18+

	Parity Transition					
	1→2	2→3	3→4	4→5	5→6	6→7
Panel A: Child Poverty (OPM)						
Mothers 30+	0.031	0.037	0.043	0.038	0.079	0.065
	(0.004)	(0.004)	(0.009)	(0.016)	(0.037)	(0.063)
	[< .001]	[< .001]	[< .001]	[.017]	[.033]	[.306]
Mothers 18+	0.038	0.048	0.051	0.053	0.065	0.034
	(0.004)	(0.004)	(0.008)	(0.015)	(0.035)	(0.062)
	[< .001]	[< .001]	[< .001]	[< .001]	[.062]	[.584]
Panel B: Log Poverty Threshold (OPM)						
Mothers 30+	0.232	0.188	0.147	0.149	0.198	0.241
	(0.002)	(0.002)	(0.003)	(0.006)	(0.015)	(0.020)
	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]
Mothers 18+	0.238	0.189	0.150	0.149	0.198	0.252
	(0.002)	(0.002)	(0.003)	(0.006)	(0.015)	(0.021)
	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]
Panel C: Log Family Income						
Mothers 30+	0.115	0.014	−0.011	0.039	−0.154	0.043
	(0.010)	(0.010)	(0.020)	(0.032)	(0.092)	(0.128)
	[< .001]	[.173]	[.595]	[.225]	[.093]	[.738]
Mothers 18+	0.107	0.002	−0.017	0.009	−0.117	0.084
	(0.009)	(0.010)	(0.020)	(0.034)	(0.086)	(0.125)
	[< .001]	[.807]	[.391]	[.798]	[.170]	[.503]
Obs. (30+)	2,644,308	1,511,086	507,798	146,809	42,508	14,887
Obs. (18+)	3,060,571	1,705,150	569,743	162,593	46,141	15,677

Notes: 2SLS estimates on the pooled ACS 2015–2024 sample. Each column instruments an indicator for having at least $n+1$ children with a twin birth at pregnancy n. The 30+ rows correspond to the baseline IV specification used in Table 2 (main text); the 18+ rows widen the sample to all mothers in the analysis age range (ages 18–59). Controls and other specification details are as in Table 2 (main text).

C.7 Mother Labor-Supply Outcomes

The fixed-resource decomposition is conservative if additional children reduce parental labor supply. To assess this, Table C5 reports twin-IV estimates for four additional labor-supply outcomes from the pooled 2015–2024 ACS: employment (indicator), usual hours per week (zero if not working), part-year work (1–39 weeks), and full-year work (40+ weeks). Each outcome is constructed from the mother's IPUMS record (`empstat`, `uhrswork`, `wkswork2`), attached to the child record via `momloc`. The specification is otherwise identical to Table C2. The pattern across rows suggests a 1–3 pp drop in employment at the lower parities, a small shift from full-year to part-year work, and modest reductions in usual hours. This suggests that additional children appear to modestly reduce labor supply (or at least, not increase it), which supports interpreting the fixed-resource decomposition as conservative ($\Delta \log R \leq 0$).

Table C5: Twin-IV estimates: mother labor-supply outcomes, ACS 2015–2024

	Parity Transition					
	1→2	2→3	3→4	4→5	5→6	6→7
Employed	−0.014	−0.028	−0.031	−0.027	0.023	−0.008
	(0.005)	(0.006)	(0.010)	(0.019)	(0.042)	(0.070)
	[.009]	[< .001]	[.003]	[.157]	[.586]	[.906]
Usual hours/week	−0.33	−0.28	0.04	−0.48	−0.86	−0.37
	(0.15)	(0.16)	(0.32)	(0.64)	(1.18)	(2.15)
	[.029]	[.092]	[.905]	[.447]	[.465]	[.865]
Part-year (1–39 wks)	0.008	0.006	0.011	0.002	0.033	−0.016
	(0.004)	(0.005)	(0.010)	(0.020)	(0.048)	(0.076)
	[.078]	[.200]	[.255]	[.910]	[.499]	[.837]
Full-year (40+ wks)	−0.008	−0.006	−0.011	−0.002	−0.033	0.016
	(0.004)	(0.005)	(0.010)	(0.020)	(0.048)	(0.076)
	[.078]	[.200]	[.255]	[.910]	[.499]	[.837]
Observations	2,644,308	1,511,086	507,798	146,809	42,508	14,887

Notes: 2SLS estimates of the effect of crossing each parity threshold on mother-level labor-supply outcomes. The endogenous regressor is an indicator for having at least $n+1$ children, instrumented with a twin birth at pregnancy n. "Employed" equals one if the mother's `empstat` is 1 (employed); "Usual hours/week" is `uhrswork`; "Part-year" ("Full-year") equals one if the mother worked 1–39 (40+) weeks last year, coded from `wkswork2`. Controls and weights are identical to Table C2. Estimated on the pooled 2015–2024 ACS; 1960 estimates are omitted because the 1960 Census did not collect usual hours worked.

D Supporting Material for the Long-Run Analysis (1900–2023)

This appendix provides context and methodological details for the long-run results reported in the main text (Figure 3, panel b).

D.1 Data and Construction for Figure 3, Panel b

The long-run extension uses IPUMS-USA Decennial Census microdata (1900–2000) and the American Community Survey (2001–2023) (Ruggles et al. 2023). Children are defined as individuals ages 0–17 linked to a parent (using the same mother/father/guardian waterfall as the CPS analysis). Child poverty is measured using the Official Poverty Measure (OPM), which is consistently available from 1950 onward in IPUMS. Data from earlier decades are used to describe changes in the family-size distribution prior to 1950 and to project back implied changes in poverty using the 1950 distribution of family sizes and resources.

To mirror the main decomposition as closely as the OPM data permit, I implement a two-factor Shapley decomposition by decade: fertility (rank-preserving transport of the family-size distribution within marital-status cells) and marriage (IPW reweighting to match the target decade's child-weighted marriage composition). The remaining change is reported as a residual capturing market income growth and any other changes that cannot otherwise be separately identified using the data available.

D.2 Historical Context: The Fertility Transition Outside the CPS Window

Panel a of Figure D1 documents the long-run decline in the prevalence of large families and provides context for interpreting panel b of Figure 3. In 1900, more than half of chil-

dren lived in families with four or more children. By 2023, that share had fallen below 15%. Mean family size declined from 4.2 to 2.4. During the 1950s and 1960s, the baby boom temporarily reversed this trend.

D.3 Stability of the Poverty–Family Size Gradient in the Census/ACS

Panel b of Figure D1 summarizes the cross-sectional relationship between family size and child poverty in the Census/ACS. First, the figure shows that the gradient is large. In each year, children in larger families face higher poverty rates than children in one-child families. For example, in 2023 children in four-child families have poverty rates roughly twice those of one-child families, and children in families with seven or more face rates 3–4 times as high. Second, this gradient is persistent. It was steepest in 1950, when overall child poverty was highest, but since 1980 the poverty gaps associated with larger family sizes have remained steep and of similar magnitude.

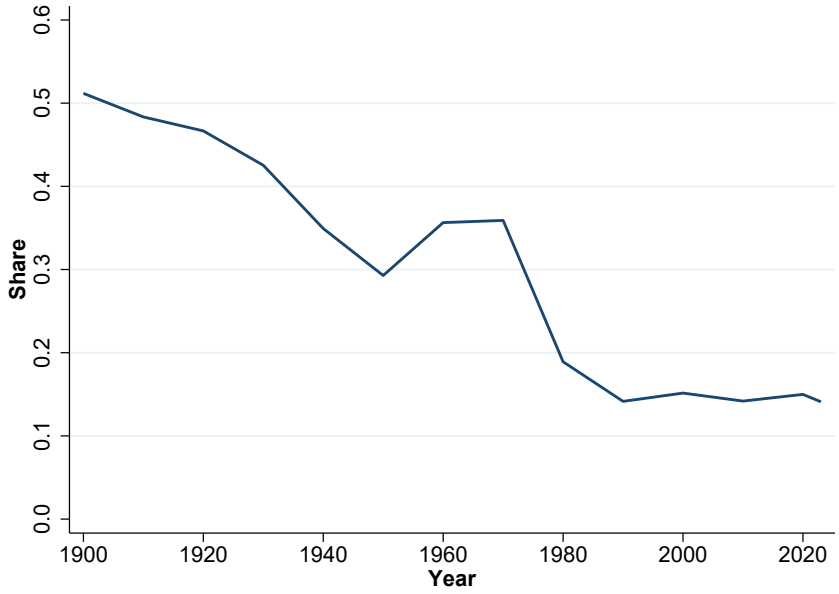
E Data Appendix

This appendix summarizes data sources, sample construction, and key variables.

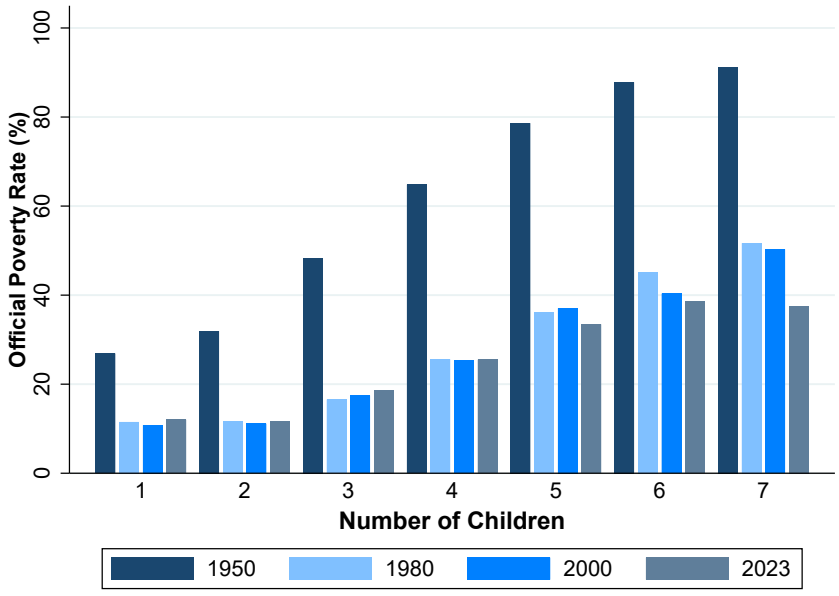
E.1 SPM Data (1967–2024)

The main analysis uses the Historical Supplemental Poverty Measure (SPM), constructed by the Center on Poverty and Social Policy at Columbia University and linked to the CPS Annual Social and Economic Supplement (Waldfogel et al. 2017). I use CPS survey years 1968–2025, corresponding to calendar years 1967–2024.

The SPM measures family resources as cash income plus in-kind transfers and refundable tax credits minus taxes and necessary expenses (including work expenses and medical out-



a. Share of children in large families (4+ children), 1900–2023



b. OPM child poverty rate by number of children, 1950–2023 (selected decades)

Figure D1: Long-run fertility and the poverty–family size gradient, 1900–2023

Notes: Panel a: share of children ages 0–17 living in families with four or more children. Panel b: OPM child poverty rate among children ages 0–17 by number of co-resident children. Both panels weighted by IPUMS person weights. Source: IPUMS USA Decennial Census and ACS, 1900–2023.

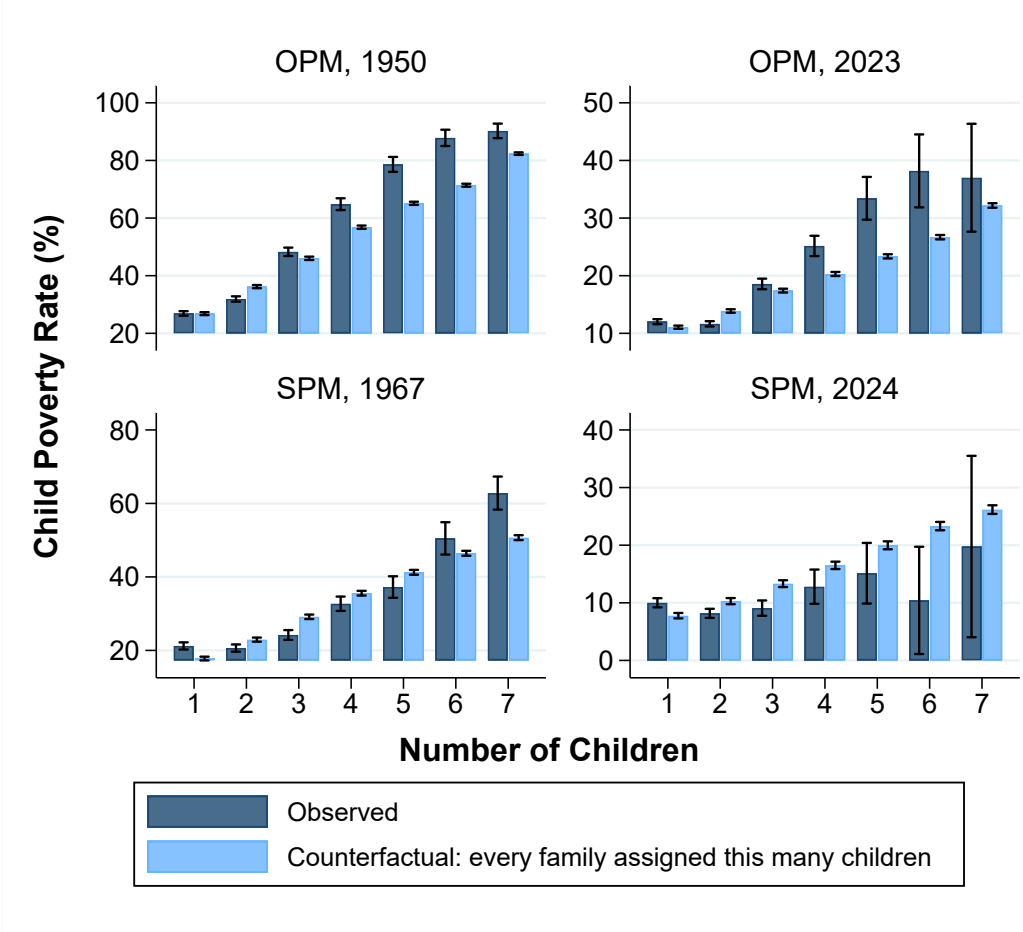


Figure D2: The poverty–family size gradient: observed and counterfactual, 1950–2024

Notes: Dark bars show the poverty rate among children living in families with the indicated number of children. Light bars show the rate that would prevail if every family with children had that number of children, holding each family's resources and number of adults fixed and rescaling its poverty threshold accordingly: the official threshold matrix by family size and number of children in the top panels (OPM; Census/ACS 1950 and 2023), and the SPM three-parameter equivalence scale in the bottom panels (SPM; CPS ASEC 1967 and 2024). Families with seven or more children are grouped at seven. Error bars are 95% confidence intervals. Weighted by IPUMS person weights (Census/ACS) and SPM unit weights (CPS ASEC).

of-pocket). SPM poverty thresholds are adjusted for family size using the Betson three-parameter equivalence scale and for geographic variation in housing costs.

The analysis sample consists of all children ages 0–17 in SPM resource units containing at least one adult and one child. Each child is linked to a parent using the following waterfall: IPUMS mother link (if age at birth is plausible), then IPUMS father link, then a guardian identified as the best-available adult in the same SPM unit. I exclude observations with missing or invalid SPM poverty status or resources. The resulting CPS–SPM analysis sample contains approximately 2.8 million child-year observations from 1967–2024.

E.2 Census and ACS Data (1900–2024)

The long-run analysis uses IPUMS-USA Decennial Census and American Community Survey microdata (Ruggles et al. 2023). For each Census/ACS year, I use the 0.5% IPUMS sample, yielding about 3.9 million child observations spanning 1900–2023.

The twin-birth IV analysis uses larger samples to provide statistical power: a 5% extract from the 1960 Census and pooled 1% ACS samples from 2015–2024. The IV sample in the main text is restricted to mothers aged 30 and older. Twin births are identified as instances in which two children linked to the same mother share the same birth year and quarter. Triplets and higher-order multiples are excluded. The IV design is parity-specific. At each margin n, I instrument an indicator for having at least $n+1$ children with a twin birth at pregnancy n, restricting the sample to mothers with at least n pregnancies.

E.3 Summary Statistics

Table E1: Summary statistics: children in CPS ASEC

	1967		2024	
	Mean	SD	Mean	SD
Demographics				
Number of children	3.29	1.68	2.35	1.19
Parent's age at first birth	23.8	5.7	27.6	6.5
Parent married (binary)	0.864	0.342	0.673	0.469
Parent's age (years)	35.3	8.5	39.0	8.6
Economic outcomes				
Family income (2024 $1000s)	50.1	32.9	111.1	104.5
Income per child (2024 $1000s)	20.5	23.5	60.2	69.3
SPM poverty (binary)	0.297	0.457	0.096	0.295
Sample size (children)	55,168		33,995	
Weighted children (millions)	71.1		68.6	

Notes: Child-weighted means and standard deviations. Sample includes all children under 18. Parent identified via IPUMS mother link, father link, or guardian cascade. All monetary values in 2024 dollars (PCE deflator). Family income is SPM total resources. Source: Historical SPM, CPS ASEC.

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